

Question	Marking Guidance	Mark	Comments
08.1	EITHER 1. The radioactively labelled carbon is converted into sugar/organic substances during photosynthesis; 2. Mass flow/translocation in the phloem throughout the plant only in plants that were untreated/B/control OR Movement of sugar/organic substances in the phloem throughout the plant only in plants that were untreated/B/control; OR 3. Movement in phloem requires living cells/respiration/active transport/ATP; 4. Heat treatment damages living cells so transport in the phloem throughout the plant only in plants that were untreated/B/control OR Heat treatment stops respiration/active transport/ATP production so transport in the phloem throughout the plant only in plants that were untreated/B/control;	2	Do not mix and match – award either mp1 and mp2 or mp3 and mp4. 1. For 'organic substance' accept named organic substance, eg glucose, sucrose, amino acid. 2. Accept 'translocation/mass transport in the phloem past the heat treatment only in the untreated plant/B/control'. 2. Accept converse for heat-treated plant/A ie Movement of sugar/organic substances/mass flow/translocation in the phloem stops (beyond the heat treatment) in treated plants/A.
08.2	1. (The water content of the leaves was) not different because (means $\pm$ 2) standard deviations overlap; 2. Water is (therefore) still being transported in the xylem (to the leaf) OR Movement in xylem is passive so unaffected by heat treatment;	2	1. For 'not different' accept 'difference is not significant' or 'difference due to chance'.

